Spanner

Software Requirements Specification, Feasibility Study & Risk Register

ITC327W Assessment 1 — Phase 1: Planning, Requirements and Feasibility (Units 1–2)

Riese Auto-Clinic — Stakeholder: Michael Riese, Owner/Head Mechanic

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Draft compiled from group chat content, the group's shared Google Doc, and the Assessment 1 brief. Amber "Assumption/flag" notes mark places where a role or detail was inferred rather than confirmed by the group — please correct these before submission. The Microsoft Project schedule is being handled separately and is referenced only at a high level here.

1. Software Requirements Specification

1.1 Project Introduction

Spanner (working title) is a workshop management system for Riese Auto-Clinic, a mechanic shop, combining a Flutter mobile application and an ASP.NET web application on a shared Supabase backend. It allows shop staff to manage parts/stock, book in vehicles and assign jobs, capture labour, and generate customer-ready PDF invoices — replacing the shop's current manual, paper- and memory-based process with a single, synchronised system usable from the workshop floor (mobile) or the office (web).

1.2 Stakeholder Profile and Current Process

Organisation: Riese Auto-Clinic, a mechanic shop. Direct stakeholder: Michael Riese, Owner/Head Mechanic, selected for his direct authority over how the shop operates and his ability to give first-hand input into system requirements. Affected users: Michael Riese and the shop's mechanics/admin staff, who would use the system directly to book vehicles, manage stock and generate invoices; shop customers are indirectly affected as recipients of the resulting invoice.

Current process: vehicles are booked in verbally, through direct communication with admin staff or the owner. Once a vehicle arrives, mechanics begin diagnostics immediately to identify faults or service points. When parts are needed, a trip is made to a local supplier to source them — sometimes across multiple stores if one does not stock everything required — before returning to install them and notify the customer of completion or complications. Once work is finished, payment is calculated by combining parts cost, a markup, and labour time, and the job is signed off. The shop keeps customer contact details along with vehicle-identifying information — odometer reading, licence plate and VIN — specifically to protect the business in case a customer becomes non-compliant.

Evidence: informal stakeholder discussion conducted directly with Michael Riese (a personal contact of a group member), documented in the group's shared planning document. A formal, scheduled stakeholder interview (Unit 1 Activity 1.2) is in progress to confirm and extend this record — see Section 1.6 and the Requirements Change Log.

1.3 Problem Statement

Current situation: bookings, parts sourcing, labour tracking and invoicing are all handled manually, relying on verbal communication, physical trips to suppliers and the owner's memory or paper notes for cost and stock information.

Problem: manually booking in a vehicle and then sourcing parts from multiple locations — when a single supplier does not carry everything needed — creates unnecessary time delays on jobs. There is no system to track where components were bought for future reference, how much stock of particular parts is held, or labour time spent per vehicle in a structured way.

Consequences: jobs take longer to complete than necessary, restocking decisions are made without good information, and producing an accurate, professional invoice at the end of a job is slower and more error-prone than it needs to be.

Need for improvement: the shop needs a system that documents each job as it comes in, tracks where parts are sourced for future purchasing decisions, maintains a record of stocked parts/service kits, tracks labour time per vehicle, and produces a clear invoice itemising component costs and labour at the end of the job.

1.4 Project Aim and Objectives

Aim: to design and develop an integrated mobile and web-based workshop management system for Riese Auto-Clinic that allows the business to manage parts/stock, book in vehicles and assign jobs, capture labour, and generate customer-ready invoices from a single, synchronised platform.

Objectives:

- Enable staff to record and manage parts/stock inventory, including where a part was bought, its cost price, and the markup/sell price charged to the customer.
- Allow staff to book in a vehicle — capturing customer contact details, VIN, licence plate and odometer reading — and assign one or more jobs to it (e.g. service, engine, suspension).
- Allow staff to capture labour hours worked on a job, together with an hourly labour rate, once the job is marked as finished.
- Automatically calculate the total job cost (parts + labour) when a job is completed.
- Generate a PDF invoice/quote for the completed job that can be sent to the customer via WhatsApp or email, or printed.
- Synchronise data in real time between the Flutter mobile app and the ASP.NET web app using a shared Supabase backend.
- Keep parts sourcing information (supplier, cost) on record to support future purchasing decisions and reduce delays from re-sourcing parts.

1.5 Project Scope (initial)

In scope: a Flutter mobile application and an ASP.NET web application sharing one Supabase backend; parts/stock management (source, cost price, sell price/markup); vehicle booking and job assignment; labour hour and rate capture; automatic job costing; PDF invoice/quote generation shareable via WhatsApp, email or print.

Out of scope: online payment processing; automated customer appointment reminders; support for multiple branches/franchise locations; a customer-facing login portal.

Acceptance criteria: both the mobile and web versions must clearly log the work done on a job and be able to export a PDF invoice for it.

Constraints: the project must be completed within a single semester, with no monetary cost involved in developing it.

Assumptions: the stakeholder (Michael Riese) will remain available for feedback and confirmation throughout the project.

1.6 Stakeholder Needs

The following needs were confirmed directly with Michael Riese through the formal stakeholder interview conducted for Activity 1.2 (transcript below): a combined mobile and web system that keeps track of stock, where parts were sourced and their costs; the same stock visible to all mechanics and to admin, so mechanics know what's available and admin can track money spent per job; a central system linking parts and labour to the specific job and vehicle they relate to, rather than tracking them separately; each mechanic able to log their own labour time against a job, so admin has a clear record of who did what and for how long when preparing an invoice; and an invoice generated automatically once a job is finished, including both labour and parts cost.

Stakeholder Interview Transcript (Activity 1.2)

Q1 So you would like us to make your life a bit easier, how can we do that for you?

A1: We would like an app for mobile and a desktop/web version that can keep track of all the stock we keep, where we got the parts, and their costs.

Q2 Okay, that makes sense. And I would assume you would want the app and web version to talk to each other and keep track of the same stock and things?

A2: Yes. We'd like all the mechanics to be able to see what we have available, what needs to get bought, and so on, but also so that the admin personnel can also keep track of the money spent on each job that comes in.

Q3 Alright, so we need to develop a back end database that tracks parts, and for which jobs they're being used for. Might I also make a suggestion? If we are tracking the parts, why not track what types of jobs is being performed on different vehicles? Have a central tracking/management system?

A3: That would be perfect. Having admin and the mechanics all stay in the loop and on the same page with what is being done and the costs involved with it as well.

Q4 That is what I had in mind as well. We can even make it so that every mechanic working on a vehicle is able to log their labour time on the app, admin can see it, and when developing the invoice, have a clear track of who did what on which vehicle and for how long, if that is what would help as well?

A4: That would honestly help us a lot with paperwork and unnecessary admin work.

Q5 Is there anything else we can do for you or that we can include into this system that would improve your situation?

A5: Yes, actually, if it would be possible, to have the system itself export an invoice after all the work is done on a vehicle, which includes all the labour and cost price.

Assumption/flag: The signed stakeholder confirmation letter referenced as evidence for this interview is being collected separately and should be attached as a supporting appendix once signed.

1.7 Functional Requirements

FR-01 The system shall allow staff to register a new vehicle booking with customer contact details, VIN, licence plate and odometer reading.

FR-02 The system shall allow staff to assign one or more job types (e.g. service, engine, suspension) to a booked vehicle.

FR-03 The system shall allow staff to record parts/stock items, including supplier, purchase (cost) price and sell price/markup.

FR-04 The system shall allow staff to link parts used to a specific job on a specific vehicle.

FR-05 The system shall allow staff to capture labour hours worked and an hourly labour rate for a job.

FR-06 The system shall automatically calculate the total job cost (parts + labour) once a job is marked complete.

FR-07 The system shall generate a PDF invoice/quote summarising parts, labour and total cost for a completed job.

FR-08 The system shall allow a generated invoice PDF to be shared via WhatsApp or email, or printed.

FR-09 The system shall synchronise stock, booking and job data between the mobile app and web app in real time via Supabase.

FR-10 The system shall allow authorised staff to view current stock levels of parts/service kits.

FR-11 The system shall maintain a per-vehicle history of past jobs, searchable by VIN or licence plate.

FR-12 The system shall restrict access to shop data to authenticated staff accounts only.

1.8 Non-Functional Requirements

NFR-01 (Usability) The interface must be simple enough for mechanics with limited software experience to use with minimal training.

NFR-02 (Performance) Core actions (booking a vehicle, adding a part, generating an invoice) should complete in a few seconds under normal connectivity.

NFR-03 (Availability) The mobile app should tolerate brief connectivity loss without losing unsaved job data, given real connectivity gaps already experienced by the group.

NFR-04 (Security) Customer and vehicle data must be protected via authenticated access and Supabase Row Level Security, and must never be committed to the public GitHub repository.

NFR-05 (Data integrity) Financial calculations (cost, markup, totals) must be accurate and auditable.

NFR-06 (Compatibility) The mobile app must run on common Android/iOS devices used by shop staff.

NFR-07 (Maintainability) The codebase must be organised and documented in GitHub to support future extension beyond this project.

1.9 Integration Requirements

The Flutter mobile app and the ASP.NET web app both connect to a single shared Supabase project (PostgreSQL database, Auth, Storage and Realtime) as the single source of truth for stock, bookings, jobs and invoices. Authentication is handled through Supabase Auth and shared across both clients so staff use one account regardless of which app they open. Changes made on one client (e.g. a new stock entry from the mobile app) propagate to the other (e.g. the web dashboard) via Supabase Realtime subscriptions. PDF invoice generation is triggered on job completion and shared through the device's native share sheet (mobile) or a direct download/share action (web), avoiding the cost and complexity of a dedicated messaging API.

1.10 Requirements Change Log

Change	What Changed	Reason
1	Web application platform specified as ASP.NET rather than a generic web app.	Aligned to the ASP.NET + Supabase technical requirement set out in the Assessment 1 brief.
2	Customer-facing login portal removed from scope.	Operational feasibility: the direct users are shop staff, not customers; economic/schedule feasibility favoured a tighter scope.
3	WhatsApp/email sharing implemented via native device share sheet rather than a paid messaging API.	Economic feasibility: keeps the project at zero development cost and avoids third-party API complexity.
4	Multi-branch/franchise support explicitly excluded.	Riese Auto-Clinic is a single-location business; out of scope was confirmed with the stakeholder relationship in mind.

1.11 Refined Project Scope

Following the stakeholder interview above and the feasibility and risk analysis in Sections 2 and 3, the group finalised the scope below to carry forward into system design.

Project Title: Spanner (RT)

Project Aim: Develop an integrated mobile and web system that allows Riese Auto-Clinic to book vehicles, assign and track jobs, manage parts stock and purchase history, log labour time per mechanic, and generate detailed invoices — all from a shared central system.

Objectives

- Provide mechanics with a simple way to view available stock, log labour time, and record parts used on jobs.
- Allow admin staff to track costs, monitor job progress, and manage stock levels.
- Maintain a single shared source of truth for jobs, parts, labour and costs across both the mobile and web applications.
- Reduce paperwork and administrative effort by automatically generating clear invoices.

In Scope

- Vehicle booking and job creation.
- Assignment of jobs to mechanics.
- Logging of labour time per mechanic per job.
- Tracking of parts on each job (including supplier and cost).
- Basic stock management (quantity on hand and low-stock awareness).
- Recording of the type of work performed.
- Generation and export of a detailed invoice (PDF) showing parts costs and labour.
- Shared data between the Flutter mobile app and ASP.NET web application via Supabase.

Out of Scope

- Customer-facing portal or client login.
- Online payment processing.
- Advanced inventory forecasting or automatic reordering.

- Integration with external accounting software.
- Real-time vehicle tracking.
- Support for multiple workshop branches.

Deliverables

- Flutter mobile application.
- ASP.NET web application.
- Shared Supabase backend and database.
- Ability to export a clear invoice PDF.

Constraints

- Must be completed within a single semester.
- No monetary costs for development tools or services (free tiers only).
- Prototype quality is acceptable (not a fully production-ready system).

Assumptions

- The stakeholder (Michael Riese) and relevant staff will remain available for feedback and meetings throughout the project.
- Mechanics and admin staff will have access to smartphones and computers.
- Basic internet connectivity will be available at the workshop.

2. Feasibility Study

Area	Evaluation
Technical	Flutter (mobile) and Supabase (backend: Postgres, Auth, Storage, Realtime) are already in use, with a working Supabase project and an in-progress mobile prototype giving the group real technical traction. The web client will use ASP.NET, integrating with the same Supabase backend. The main technical gap is the group's limited prior ASP.NET experience, which is a manageable learning curve given the semester timeframe and Supabase handling most backend logic. PDF generation and native share-to-WhatsApp/email are both achievable with free, well-documented libraries. Conclusion: technically feasible, with ASP.NET familiarisation flagged as a schedule risk (see Risk Register).
Operational	The proposed workflow (book a vehicle, assign a job, capture parts and labour, generate an invoice) maps directly onto Riese Auto-Clinic's actual process as described by Michael Riese, so no retraining in a new way of working is required — only digitising steps staff already perform. The main operational requirement is a simple, low-friction interface, since the direct users are mechanics and admin staff rather than technical staff. Conclusion: operationally feasible provided the interface stays simple (see NFR-01).
Economic	The project uses free-tier tooling throughout: Flutter and ASP.NET are free and open-source, Supabase's free tier covers a project of this scale, and GitHub is free for the group's use. Sharing invoices uses the device's native share sheet rather than a paid WhatsApp Business API or SMS gateway. This matches the stated project constraint of zero development budget. Conclusion: economically feasible at no direct cost to the group or the stakeholder.
Schedule	The group has a head start (mobile prototype and Supabase project already created before the module formally began) and a single-semester deadline. The realistic risk is the web application (ASP.NET) and full mobile-web-Supabase integration, which have not yet started. Conclusion: feasible only if the remaining scope is kept tight — the refined scope below deliberately excludes lower-priority features to protect the schedule.
Legal	The system stores personal customer contact details and vehicle-identifying data (VIN, licence plate, odometer) that Riese Auto-Clinic already collects for its own compliance protection. This data must be handled in line with South Africa's Protection of Personal Information Act (POPIA) — collected only for the stated business purpose, access-restricted to authenticated staff via Supabase Row Level Security, and never committed to the public GitHub repository. Conclusion: legally feasible provided access control and data-minimisation practices in NFR-04 are implemented.
Ethical	The system must represent parts costs and labour charges to customers accurately and transparently, since the invoice is the customer-facing record of what they are charged. Stakeholder and customer information must only be used for the stated project and assessment purpose. Conclusion: ethically feasible with no material concerns identified, provided invoice calculations remain accurate and auditable (see NFR-05).

Overall Feasibility Decision

The project is feasible with changes. It is well supported technically and economically by the group's existing Supabase project and mobile prototype, and it is operationally and ethically sound because it digitises a process the stakeholder already follows. The two conditions attached to this decision are: (1) the ASP.NET web application must start in parallel with continued mobile work rather than after it, given the group's limited prior ASP.NET experience (see R-03), and (2) the scope must stay to the refined list in Section 1.11 to protect the single-semester schedule.

3. Project Risk Register

Likelihood and Impact are rated Low / Medium / High (1–3); Score = Likelihood × Impact, bucketed as Low (1–2), Medium (3–4) or High (6–9).

ID	Risk Statement	Category	Likelihood	Impact	Score / Level	Mitigation	Contingency	Owner	Status
R-01	Michael Riese's availability for feedback and confirmation reduces as the semester progresses, delaying requirement sign-off.	Stakeholder	Medium	High	6 / High	Agree a light-touch, low-effort check-in channel (WhatsApp) early, and batch questions to minimise his time burden.	Fall back on previously confirmed requirements and document any assumption explicitly for later confirmation.	Gary	Open
R-02	The stakeholder requests additional features once he sees the working prototype, expanding scope beyond what the semester allows.	Scope	Medium	Medium	4 / Medium	Maintain a clearly refined, agreed scope (below) and log any new request in the requirements change log rather than building it immediately.	Park new requests as a 'Phase 2 / future work' list to revisit after submission.	Denny	Open
R-03	The web application (ASP.NET) has not yet started and the group has limited prior ASP.NET experience, risking a late or rushed build.	Technical	Medium	High	6 / High	Start ASP.NET scaffolding and the Supabase connection early in Phase 2, in parallel with mobile work rather than after it.	Reduce the web app to a read-only dashboard of the same Supabase data if full parity can't be reached in time.	Khumo	Open
R-04	The project outgrows Supabase's free-tier storage/bandwidth limits before the project is demonstrated.	Technical / Infrastructure	Low	Medium	2 / Low	Monitor usage via the Supabase dashboard monthly; avoid storing large files (e.g., full-resolution photos) unnecessarily.	Compress or offload large assets, or apply for Supabase's student/hobby tier if a limit is approached.	Gary	Open
R-05	Real-time sync between the mobile app and web app produces conflicting or duplicate records (e.g., two staff editing the same job).	Technical / Integration	Low	Medium	2 / Low	Use Supabase's row-level timestamps and design simple, single-writer-per-record flows where possible.	Add a manual 'last updated by' field so conflicts are visible and can be corrected by staff.	Gary	Open
R-06	A group member loses connectivity or availability at short notice (already experienced once this phase), delaying their task.	Team / Resource	Medium	Medium	4 / Medium	Keep the progress tracker current so gaps are visible early, and maintain a WhatsApp-first coordination habit that doesn't depend on any one channel.	Reassign or share the affected task temporarily among members with capacity.	Wisani	Monitoring
R-07	Task completion becomes uneven across members as deadlines approach, with some falling behind.	Team / Schedule	Medium	Medium	4 / Medium	Review the Task Progress tracker weekly and flag slipping tasks before they become blockers.	Redistribute the task or bring in support from a member with lighter load (as already planned for Khumo supporting Gary).	Wisani	Monitoring
R-08	The formal stakeholder interview (U1-05) and supporting evidence are not finalised in time to support the SRS.	Documentation / Evidence	Low	High	3 / Medium	Schedule the interview as a fixed WhatsApp call this week rather than leaving it open-ended, given direct access to the stakeholder.	Use the existing informal stakeholder discussion as interim evidence, clearly logged as informal pending the formal record.	Gary	Open

ID	Risk Statement	Category	Likelihood	Impact	Score / Level	Mitigation	Contingency	Owner	Status
R-09	The GitHub repository is not set up in time, leaving no evidence of individual contribution as required by the assessment.	Project Management	Low	High	3 / Medium	Kea has taken ownership of repository setup and should create it with a proper README promptly.	If delayed, migrate existing local project history into the repository as soon as it is created so contribution history is not lost.	Kea	Open
R-10	Customer and vehicle data (contact details, VIN, licence plate) is exposed through inadequate access control or accidentally committed to GitHub.	Legal / Compliance	Low	High	3 / Medium	Apply Supabase Row Level Security from the start and add real stakeholder data to .gitignore / exclude from any repository content.	Rotate any exposed credentials immediately and review repository history to remove the exposure if it occurs.	Denny	Open
R-11	PDF invoice generation does not integrate cleanly with the chosen library, producing incorrectly formatted or incomplete invoices.	Technical	Low	Medium	2 / Low	Prototype the PDF generation step early with sample data rather than leaving it until the end of development.	Fall back to a simpler, text-based PDF layout if a more polished template proves too time-consuming to implement.	Gary	Open
R-12	The Microsoft Project schedule underestimates remaining effort, causing later phases to slip.	Schedule	Medium	Medium	4 / Medium	Base task durations on actual Phase 1 pace rather than optimistic estimates, and review the schedule against real progress each phase.	Cut lower-priority scope items (see refined scope) rather than compressing testing or integration time.	Denny	Open

Three Highest Project Risks

- R-01 — Stakeholder availability declining over the semester (Score 6/High) — everything downstream depends on continued access to Michael Riese.
- R-03 — ASP.NET web app not yet started, limited prior experience (Score 6/High) — the single biggest technical gap between current progress and the full scope.
- R-12 — Microsoft Project schedule underestimates remaining effort (Score 4/Medium) — the next-highest risk, and directly relevant once the current draft schedule is reviewed against actual Phase 1 pace.

Assumption/flag: The owner for R-03 (Khumo, web app) reflects Khumo picking up ownership of a currently-unassigned area — no one has yet been confirmed to build the ASP.NET web app. Please confirm with the group rather than treating it as final. R-09 (GitHub setup) is now confirmed to Kea, who volunteered for it directly.